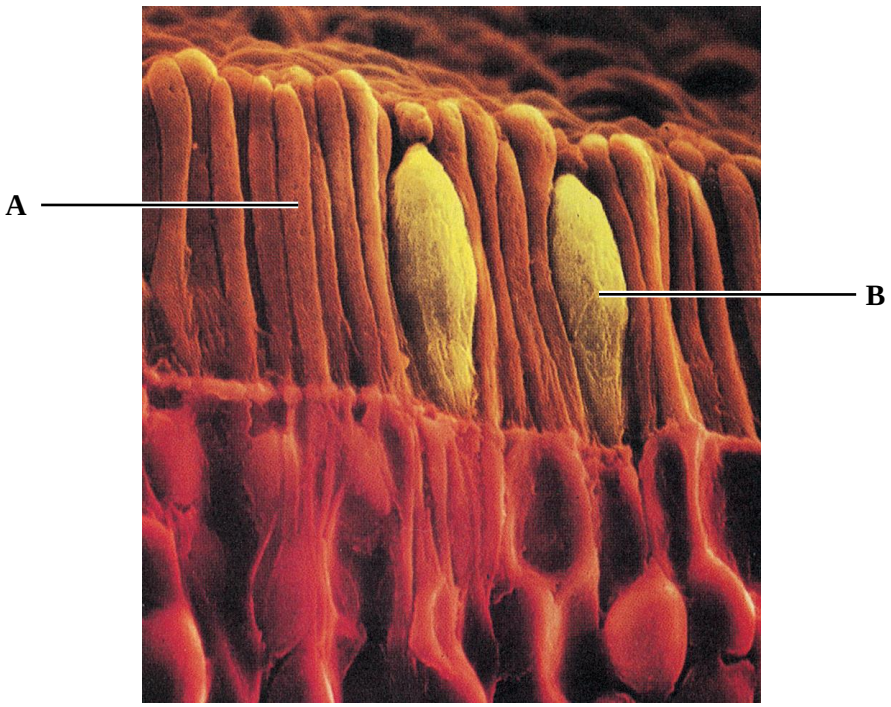
Structured questions

|!|BCELB0260000001|!| ★ **Cross topic**

The electron micrograph below shows two types of photoreceptors found on the retina of human eyes.



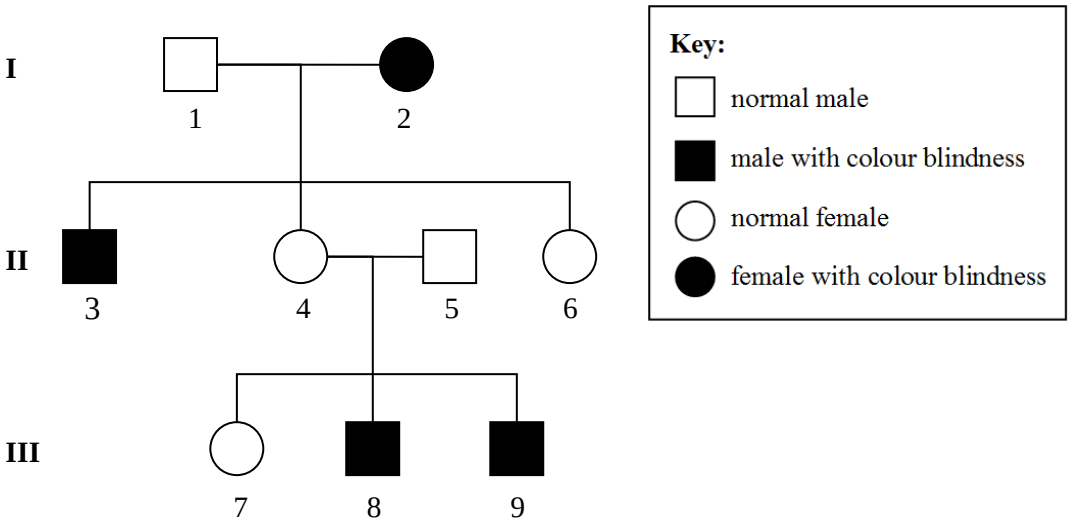
Photoreceptors and nerve fibres (×5000)

(a) Identify photoreceptors A and B. (2 marks)

(b) When an image of an object falls at the periphery of the retina, the object can be seen but the colour of the object cannot be distinguished clearly. Explain this phenomenon with reference to the functions of photoreceptors A and B. (3 marks)

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(c) Red-green colour blindness is a sex-linked recessive genetic disorder. The pedigree below shows the inheritance of red-green colour blindness in a family.



(i) Based on the pedigree, explain why red-green colour blindness is an X-linked genetic disease instead of a Y-linked genetic disease. (2 marks)

(ii) Deduce, with a reason, which individual(s) must be a carrier of red-green colour blindness. (2 marks)

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##

(a) A is a rod cell. (1)

B is a cone cell. (1)

(b) At the periphery of the retina, rod cells are abundant but only very few cone cells are present. (1)

When light falls on the periphery of the retina, many rod cells and only very few cone cells are stimulated. Rod cells are responsible for black and white vision while cone cells are responsible for colour vision. (1)

Because too few cone cells are stimulated, the colour of the object cannot be distinguished clearly. (1)

(c) (i) If red-green colour blindness is a Y-linked disease, individual 3 must have received a Y chromosome with the allele for red-green colour blindness from his father, individual 1. (1)

However, individual 1 has normal vision. Therefore, red-green colour blindness is not a Y-linked disease. (1)

(ii) Individual 2 is a female with red-green colour blindness, so she must have two alleles for red-green colour blindness on her X chromosomes. (1)

Individuals 4 and 6 are her daughters. They must have each received an allele for red-green colour blindness from individual 2. (1)

Hence, individuals 4 and 6 must be carriers of red-green colour blindness.

(9 marks)

##

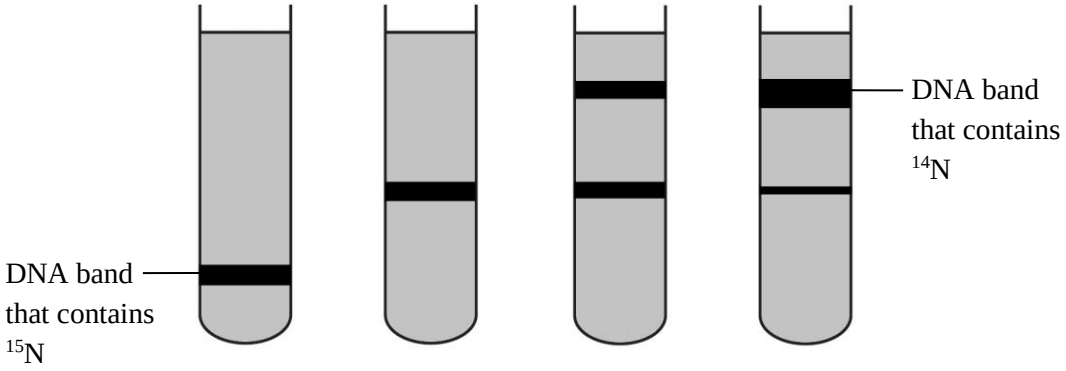
	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Applying concepts	3
(c)(i)	Analysing data	1
	Applying concepts	1
(c)(ii)	Analysing data	1
	Applying concepts	1

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|!|BCELA0260000002|!| ★★ **SI** **NOS**

In 1958, Matthew Meselson and Franklin Stahl carried out an experiment that provided evidence to support the hypothesis of semi-conservative replication of DNA.

They grew *Escherichia coli* bacteria in a culture medium that contained only the heavy isotope of nitrogen (^{15}N). The nitrogen was incorporated into the DNA of the bacteria. They then introduced the bacteria to a culture medium that contained only the light isotope of nitrogen (^{14}N). After each cell division, DNA was extracted from some of the bacteria and centrifuged to separate it. The diagram below shows the bands of DNA in different generations of the bacteria culture.



(a) Into which part of the bacterial DNA was the nitrogen incorporated? (1 mark)

(b) What is meant by the term semi-conservative replication? (2 marks)

(c) Explain why the results support the hypothesis of semi-conservative replication of DNA. (3 marks)

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(d) Give ***one*** aspect about the nature of science that can be demonstrated in each of the following events.

(i) The results of the above experiment disproved the conservative model of DNA replication. (The conservative model suggested that an entirely new molecule is synthesized from the original DNA molecule) (1 mark)

(ii) The results of the above experiment provided further support to the Watson and Crick model of DNA structure. (1 mark)

##

(a) Nitrogenous bases (or all the four named nitrogenous bases) (1)

(b) In a new DNA molecule, one strand is conserved from the original molecule, (1)
while the other strand is newly synthesized (from free nucleotides). (1)

(c) In generation 1, the results show that the new DNA strands contained both ^{15}N and ^{14}N. (1)
This suggests that the original DNA strand that contains ^{15}N was retained in the newly synthesized DNA molecule (1)
and the new DNA strand was synthesized using ^{14}N. (1)

(d) (i) Scientific knowledge is tentative and subject to change. (1)

(ii) Scientists build on the work of other scientists. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Analysing data	1
	Applying concepts	2
(d)(i)	Critical thinking	1
(d)(ii)	Critical thinking	1

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|!|BCELA0260000003|!| ★★

In fruit flies, the length of wings is controlled by a pair of alleles, an allele for long wings and another allele for vestigial wings. In a genetic experiment, three groups of fruit flies (each group consisted of individuals of the same genotype for wing length) were crossed as follows.

Cross I

Parent:	A	×	A
	(long wings)		(long wings)
F_1:	68 long wings and 32 vestigial wings		

Cross II

Parent:	A	×	B
	(long wings)		(long wings)
F_1:	all have long wings		

Cross III

Parent:	A	×	C
	(long wings)		(vestigial wings)
F_1:	56 long wings and 61 vestigial wings		

(a) Based on the above crosses, determine the dominant allele for the length of wings in fruit flies. State and explain which cross helps you find your answer. (4 marks)

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(b) Using the letter **G** to represent the dominant allele for wing length and **g** for the recessive allele for wing length,

(i) state the possible genotypes of the three groups of fruit flies. (3 marks)

(ii) construct a genetic diagram for cross III. (4 marks)

##

(a) The allele for long wings (1)

Cross I (1)

Some of the offspring in cross I have the allele for vestigial wings which must have been inherited from either of the parents. (1)

One of the parents must have both the alleles for long wings and vestigial wings, but they display the long wing character. (1)

Hence, it can be deduced that the allele for long wings is dominant over the allele for vestigial wings.

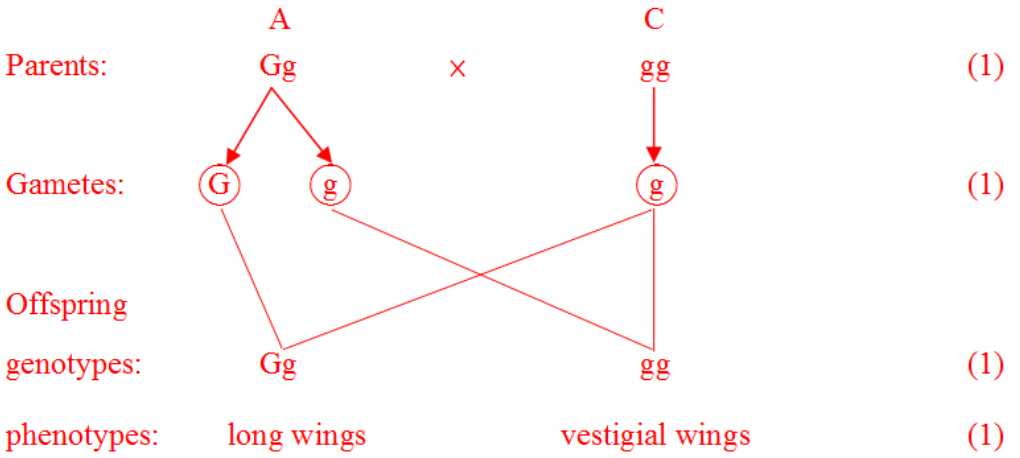
(b) (i) Group A: Gg (1)

Group B: GG (1)

Group C: gg (1)

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(ii)



Wrong format: deduct 1 mark

Wrong use of symbols: deduct 1 mark

(11 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Applying concepts	2
(b)(i)	Analysing data	3
(b)(ii)	Presenting information	4

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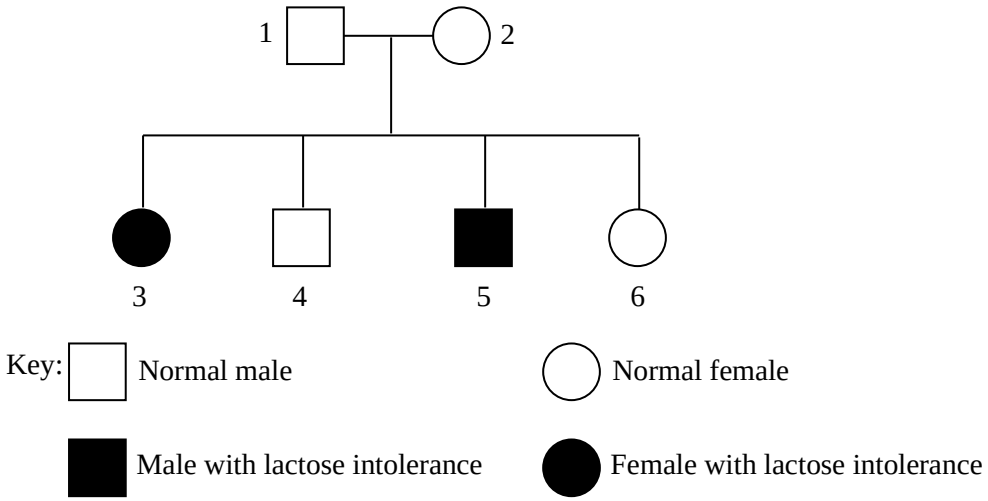
|!|BCELA0260000004|!| ★★ **Cross topic**

Lactase is an enzyme necessary for the digestion of lactose in milk.

Adults in some parts of the world rarely consume milk and often show lactose intolerance because they do not produce lactase. In contrast, adults in parts of the world where diary consumption is common do produce lactase. This is because a new allele for lactase production has arisen through gene mutation in the populations of these areas.

(a) State what is meant by gene mutation. (1 mark)

The following pedigree shows the inheritance of lactose intolerance in a family:



(b) Deduce whether the allele for the ability to produce lactase is dominant or recessive. (Note: Marks will not be awarded for genetic diagrams.) (5 marks)

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(c) Individual 5 is going to marry a woman heterozygous for the ability to produce lactase. Draw a genetic diagram to show the possible genotypes and phenotypes of their offspring. (Use L to represent the dominant allele and l to represent the recessive allele.) (3 marks)

(d) Explain why adults who cannot produce lactase may experience diarrhoea after consuming milk. (3 marks)

##

(a) A gene mutation is a change in the base sequence of the DNA in a gene. (1)

(b) Individuals 3 and 5 show lactose intolerance. They must have received at least one non-functional allele for lactase production from either of their parents (individual 1 or 2). (1)

Individuals 1 and 2 can produce lactase. Each of them must carry at least one functional allele for lactase production. (1)

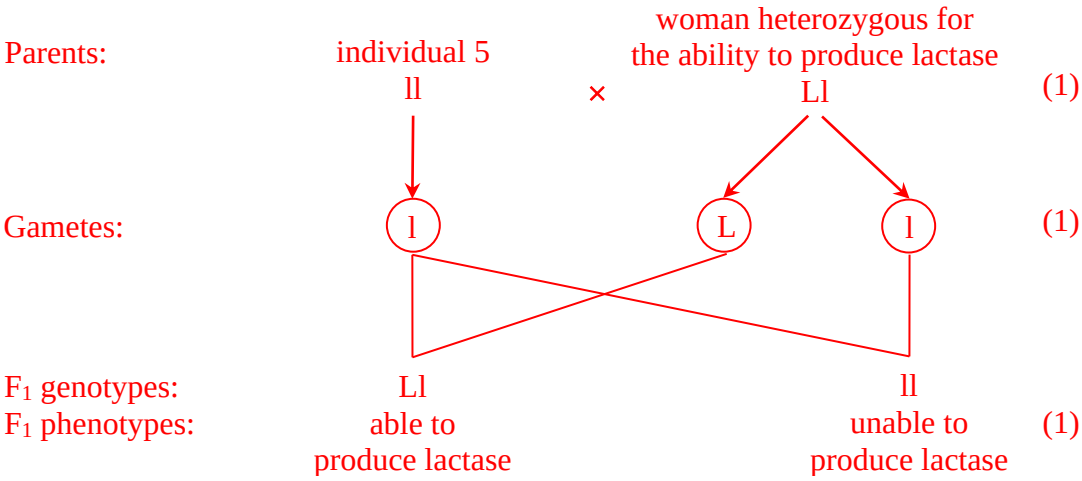
At least one of individuals 1 and 2 is heterozygous. (1)

In the heterozygous condition, the dominant allele will be expressed while the recessive allele will be masked. (1)

Thus, the allele for the ability to produce lactase in adulthood is dominant. (1)

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(c)



(d) In the absence of lactase, lactose in milk cannot be digested and absorbed, and therefore remains in the intestines. (1)

The lactose will lower the water potential the intestines. (1)

This draws water into the intestines / prevents the absorption of water in the intestines by osmosis, resulting in diarrhoea. (1)

(12 marks)

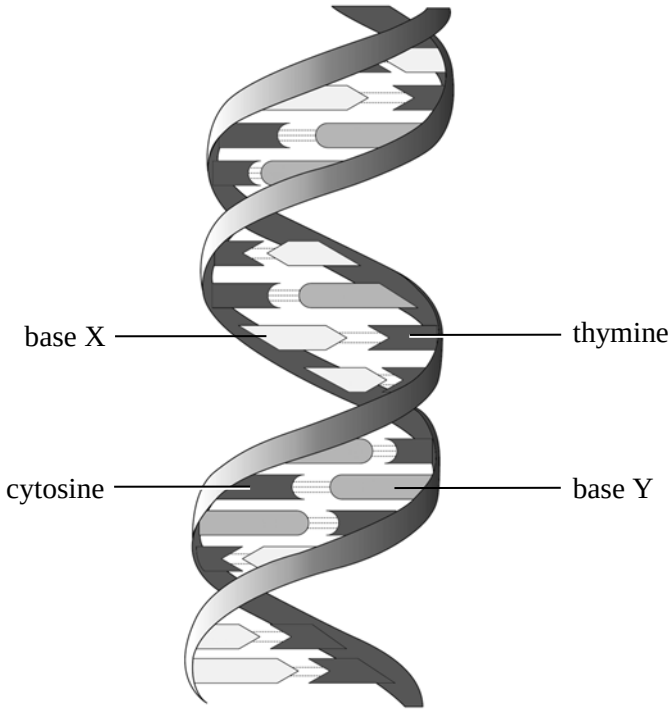
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Analysing data	2
	Applying concepts	3
(c)	Presenting information	3
(d)	Applying concepts	3

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|!|BCELC0260000005|!|

The diagram below shows a section of DNA.



(a) Name bases X and Y. (2 marks)

(b) (i) What is the structure of DNA? (1 mark)

(ii) Explain why DNA has a stable structure. (2 marks)

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(c) Describe the process of DNA replication. (4 marks)

##

(a) Base X: adenine (1)
Base Y: guanine (1)

(b) (i) Double helix (1)

(ii) The sugar-phosphate backbone of DNA is strong due to strong covalent bonds between the sugar of nucleotide and the phosphate group of the next nucleotide. (1)
The double-helical structure is maintained by many hydrogen bonds between the bases of the two DNA strands. (1)

(c) The hydrogen bonds between the bases of two DNA strands break, causing the strands to separate. (1)
Each DNA strand now acts as a template for free nucleotides to bind to by complementary base pairing. (1)
Adjacent nucleotides are joined together by DNA polymerase to form a new DNA strand. (1)
The parent strand and its new complementary DNA strand coil into a double helix. (1)

(9 marks)

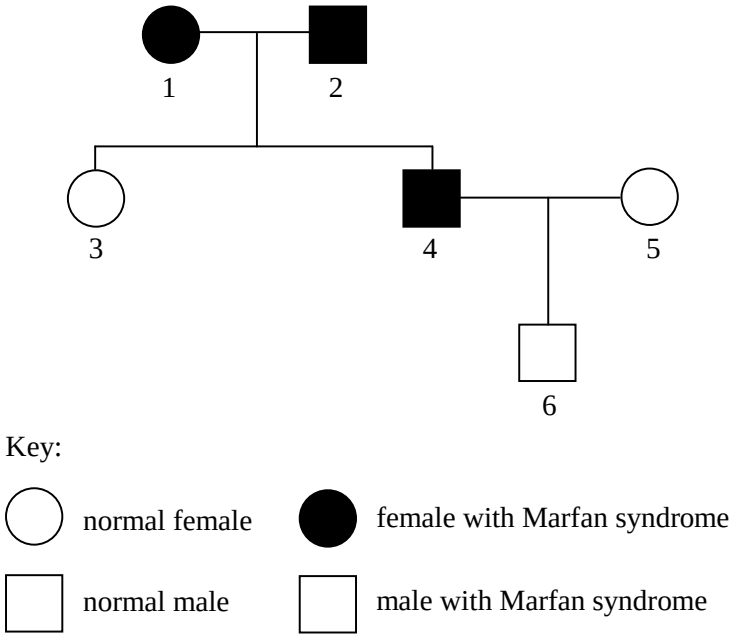
##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	2
(c)	Understanding basic concepts	4

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|!|BCELB0260000006|!| ★

Marfan syndrome is an autosomal (non-sex-linked) hereditary disease that affects the connective tissue of the body. The pedigree below shows the inheritance of Marfan syndrome in a family.



(a) Based on the pedigree above, deduce whether the allele for Marfan syndrome is dominant or recessive. (5 marks)

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(b) If individuals 4 and 5 are going to have another child, what is the probability of this child having Marfan syndrome? Illustrate your answer with a genetic diagram.

(5 marks)

##

(a) Individual 3 is normal. She must have at least one normal allele that is inherited from her parents, either individual 1 or 2. (1)

Individuals 1 and 2 have Marfan syndrome. Each of them must have at least one allele for Marfan syndrome. (1)

Hence, at least one of individuals 1 and 2 is heterozygous. (1)

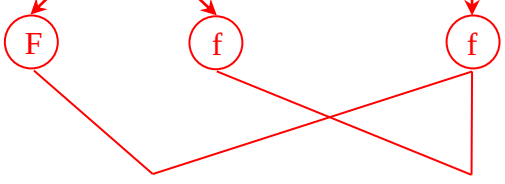
Only a dominant allele can be expressed under the heterozygous condition. (1)

As both individuals 1 and 2 have Marfan syndrome, the allele for Marfan syndrome is dominant. (1)

(b) Let F represent the allele for Marfan syndrome and f represent the normal allele. (1)

Parents: individual 4 (Ff) × individual 5 (ff) (1)

Gametes: F, f, f (1)



F_1 genotypes: Ff, ff

F_1 phenotypes: with Marfan syndrome : normal (1)

Ratio: 1 : 1

The probability of the child having Marfan syndrome is ½ (or 50%). (1)

(10 marks)

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##

	Skills assessed	Marks
(a)	Analysing data	2
	Applying concepts	3
(b)	Presenting information	4
	Mathematical skills	1

|!|BCELB0260000007|!| ★

Red-green colour blindness in humans is a genetic disorder caused by a recessive allele located on the X chromosome. People with red-green colour blindness have difficulty in distinguishing between red and green colours. It is more common in males than in females.

(a) State the name given to the inheritance pattern of red-green colour blindness.

(1 mark)

(b) Explain why red-green colour blindness is more common in males than in females.

(4 marks)

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(c) Given that the dominant allele for normal colour vision is represented by R while the recess allele for red-green colour blindness is represented by r. Using a genetic diagram, determine the probability of a colour-blind son being born to a woman who is a carrier and a man who is red-green colour-blind. (4 marks)

##

(a) Sex linkage / X linkage (1)

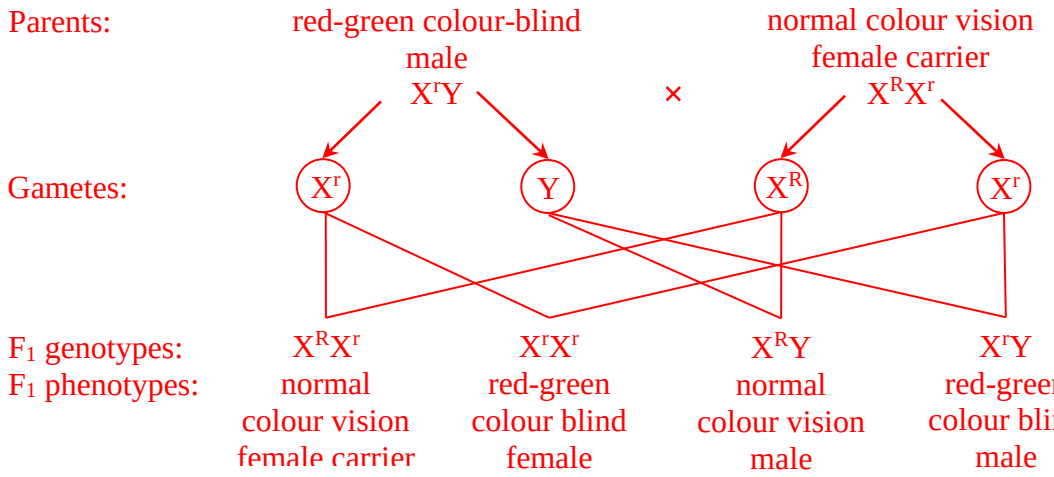
(b) Females have two X chromosomes. (1)

A female will be red-green colour-blind only if both of her X chromosomes carry the recessive allele. (1)

Males have one X chromosome and one Y chromosome. (1)

Since the gene is not found on the Y chromosome, a male having one recessive allele will be red-green colour-blind. (1)

(c)



The probability of a red-green colour-blind son being born is $^1/4$ (or 25%). (1)

(9 marks)

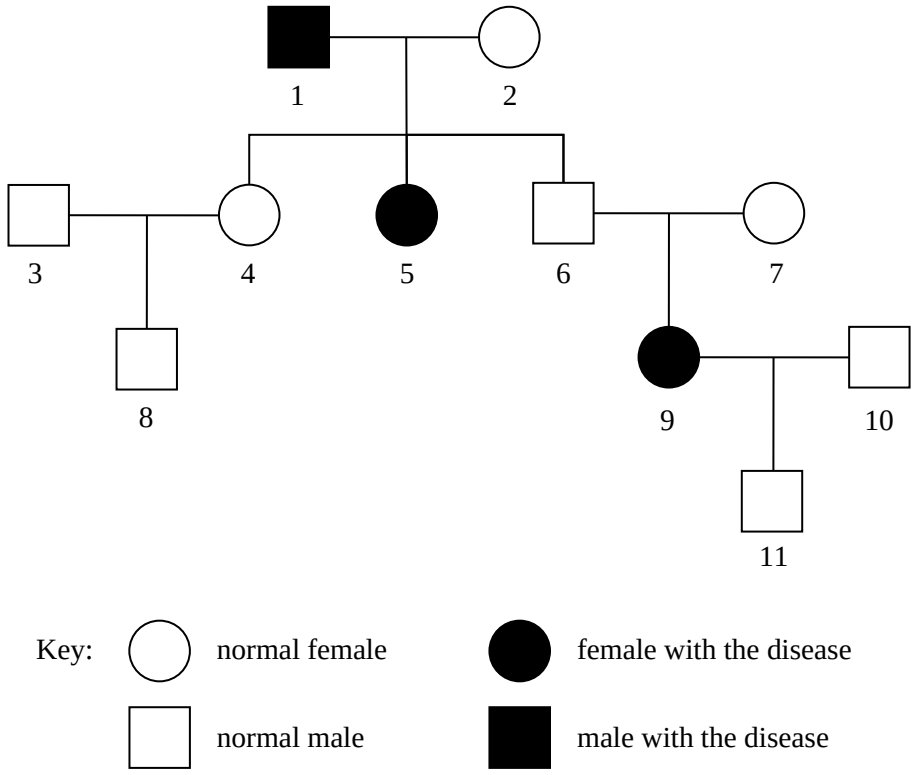
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##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	4
(c)	Presenting information	3
	Mathematical skills	1

|!|BCELC0260000008|!|

The pedigree below shows the inheritance of a disease in a family. Whether or not a person develops the disease is controlled by a pair of allele.



(a) Deduce whether the allele causing this disease is dominant or recessive. (5 marks)

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(b) Explain why the disease is ***not*** sex-linked. (7 marks)

##

(a) Individual 9 has the disease. She must have at least one allele for the disease that is inherited from her parents, either individual 6 or 7. (1)
Both individuals 6 and 7 are normal. They must have at least one normal allele. (1)
Thus, at least one of individuals 6 and 7 are heterozygous. (1)
In the heterozygous condition, the dominant allele will be expressed while the recessive allele will be masked. (1)
Thus, the allele causing the disease is recessive. (1)

(b) Females do not have Y chromosomes. (1)
However, individuals 5 and 9 have the disease and thus the disease is not Y-linked. (1)
If the recessive allele for the disease is X-linked, individual 9 having the disease must have the recessive allele for the disease on both of her X chromosomes. (1)
All of her offspring must inherit the recessive allele for the disease from her. (1)
Individual 11 is a male. He has only one X chromosome. (1)
If the allele for the disease is X-linked, it must be expressed. (1)
However, from the pedigree, individual 11 is normal. Thus, the disease is not X-linked. (1)

(12 marks)

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##

	Skills assessed	Marks
(a)	Analysing data	2
	Applying concepts	3
(b)	Analysing data	3
	Applying concepts	4

|!|BCELB0260000009|!| ★ **SI**

Two pure-breeding strains of pea plants were obtained. One strain had yellow seeds and the other had green seeds. The two strains were crossed, producing seeds (F_1 generation) that are all yellow.

(a) Using plants producing yellow seeds as an example, describe how pure-breeding plants can be obtained. (3 marks)

(b) Given that the seed colour in pea plants is controlled by a pair of allele, one allele for yellow seeds and the other for green seeds. Which of the two alleles is dominant? Explain your answer. (4 marks)

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(c) After the F_1 seeds had grew into plants, they were allowed to self-pollinate. There were yellow and green seeds in the F_2 generation. A sample of the seeds was counted. 705 were yellow seeds, and 224 were green seeds.

(i) Calculate the ratio of yellow seeds to green seeds in the F_2 generation. (1 mark)

(ii) Draw a genetic diagram to show why this ratio of yellow seeds to green seeds was produced. Use the symbol A to represent the dominant allele, and a to represent the recessive allele. (4 marks)

##

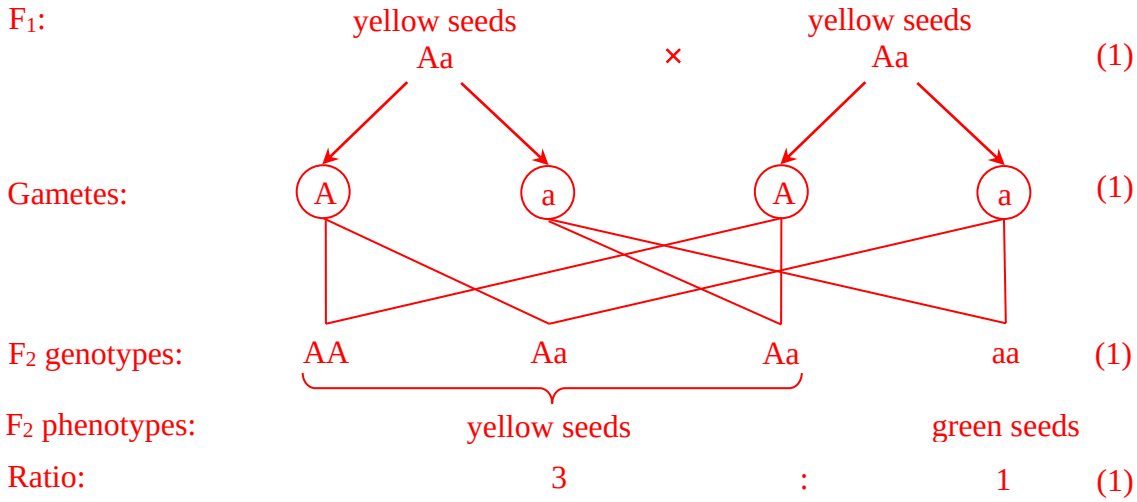
(a) Plants growing from yellow seeds are self-pollinated. (1)
After fertilization, collect the seeds produced. Select the yellow seeds for planting while discarding all the green seeds. (1)
Repeat the procedure for several generations until no more green seeds are produced. (1)

(b) Yellow seeds (1)
Since the parent plants are pure-breeding, the parent with yellow seeds carries two alleles for yellow seeds while the parent with green seeds carries two alleles for green seeds. (1)
F_1 offspring must have received an allele for yellow seeds and an allele for green seeds from their parents / must be heterozygous (1)
In the heterozygous condition, only the dominant allele is expressed. Thus, the allele for yellow seeds is dominant. (1)

(c) (i) 3.15:1 (1)

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(ii)



(12 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Applying concepts	4
(c)(i)	Mathematical skills	1
(c)(ii)	Presenting information	4

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|!|BCELC0260000010|!|

The table below shows the number of juvenile trout in different length intervals. All these trout were hatched from the same batch of eggs, and the eggs had been allowed to develop in the same controlled environment.

Length interval (cm)	Number of juvenile trout
7.0 – 7.4	0
7.5 – 7.9	4
8.0 – 8.4	10
8.5 – 8.9	16
9.0 – 9.4	34
9.5 – 9.9	80
10.0 – 10.4	100
10.5 – 10.9	108
11.0 – 11.4	68
11.5 – 11.9	40
12.0 – 12.4	16
12.5 – 12.9	12
13.0 – 13.4	8
13.5 – 13.9	4

(a) Suggest ***three*** environmental factors that would affect the length of trout. (3 marks)

_____

_____

_____

(b) Calculate the percentage of juvenile trout that are longer than 11.9 cm. Show your working. (2 marks)

_____

_____

(c) Besides the effects of environmental factors, suggest another reason why the trout show variation in length. (1 mark)

_____

_____

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(d) Which type of variation does the length of trout show? Explain your answer.

(2 marks)

_____

_____

##

(a) Any ***three*** of the following: (3)

- water temperature
- oxygen content in water
- amount of nutrients provided daily
- water salinity

(Or other reasonable answers)

(b) Percentage of juvenile trout that are longer than 11.9 cm

= (16 + 12 + 8 + 4) / 500 × 100% (1)

= 8% (1)

(c) There are genetic variations among the trout. (1)

(d) Continuous variation (1)

This is because the length of trout shows a continuous range of values with intermediates between the longest and shortest individuals. (1)

(8 marks)

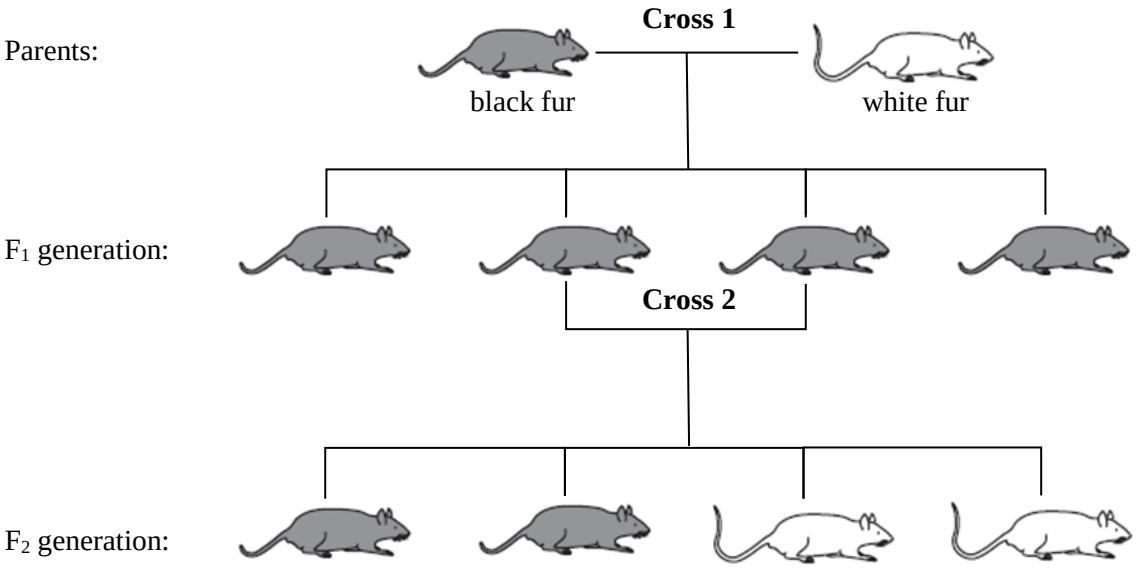
##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Mathematical skills	2
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	2

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|!|BCELB0260000011|!| ★

The fur colour of mice is controlled by a pair of alleles. The diagram below shows the inheritance of fur colour in mice.



(a) Which of the crosses allows you to deduce which fur colour is the dominant trait? (1 mark)

_____

Use F to represent the allele for the dominant trait and f to represent the allele for the recessive trait.

(b) State the genotype of

(i) the parent with white fur. (1 mark)

_____

(ii) the F_1 generation. (1 mark)

_____

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(c) (i) Draw a genetic diagram to represent cross 2. (3 marks)

(ii) What proportion of the F_2 generation is expected to have white fur? (1 mark)

(iii) The actual proportions of black and white mice in the F_2 generation are different from the proportions predicted based on Mandel's laws of inheritance. Explain why. (2 marks)

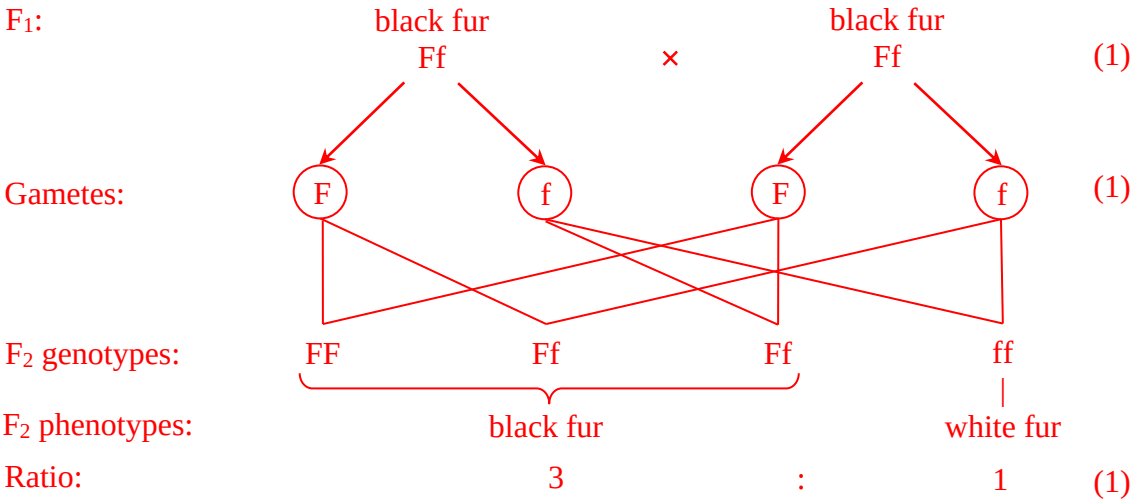
##

(a) Cross 2 (1)

(b) (i) bb (1)

(ii) Bb (1)

(c) (i)



(ii) ¼ or 25% (1)

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(iii) Fertilization occurs randomly (1)
and the number of offspring produced is small. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	1
(c)(i)	Presenting information	3
(c)(ii)	Mathematical skills	1
(c)(iii)	Critical thinking	2

|!|BCELB0260000012|!| ★ **Cross topic**

Huntington's disease is a genetic disorder that causes damage to brain cells. It is caused by a dominant allele of a gene on chromosome number 4. The symptoms usually develop in adulthood and include uncontrollable muscular movements, short-term memory loss and changes in mood.

(a) Name the part of the brain that controls muscular movements. (1 mark)

(b) Describe how the brain initiates muscular movements. (3 marks)

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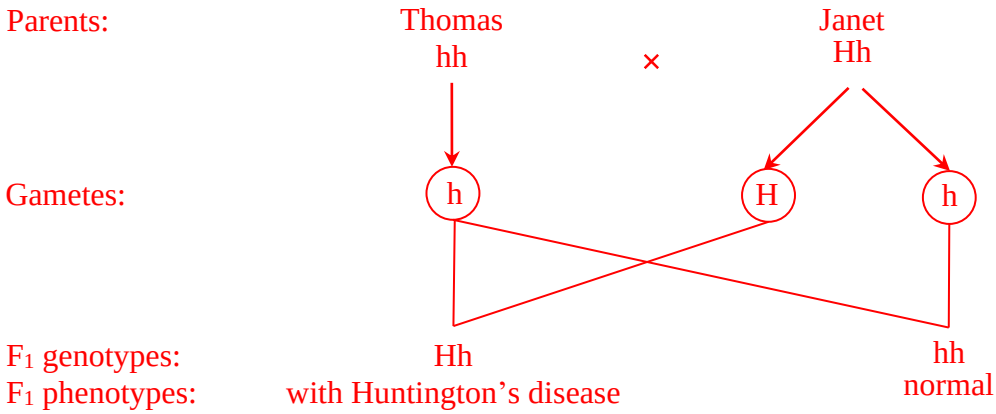
(c) Thomas and Janet married and had children. Genetic tests later revealed that Thomas was homozygous recessive for Huntington's disease but Janet was heterozygous. Determine all the possible genotypes and phenotypes of their children using a genetic diagram. (Note: Punnett square is not accepted.) (4 marks)

##

(a) Cerebrum / Cerebral cortex / Motor cortex (1)

(b) The motor cortex generates nerve impulses (1)
which are sent through motor neurones down the spinal cord (1)
to stimulate muscles to contract. (1)

(c) Let H represent the allele for Huntington's disease and h represent the normal allele. (1)



(1)

(1)

(8 marks)

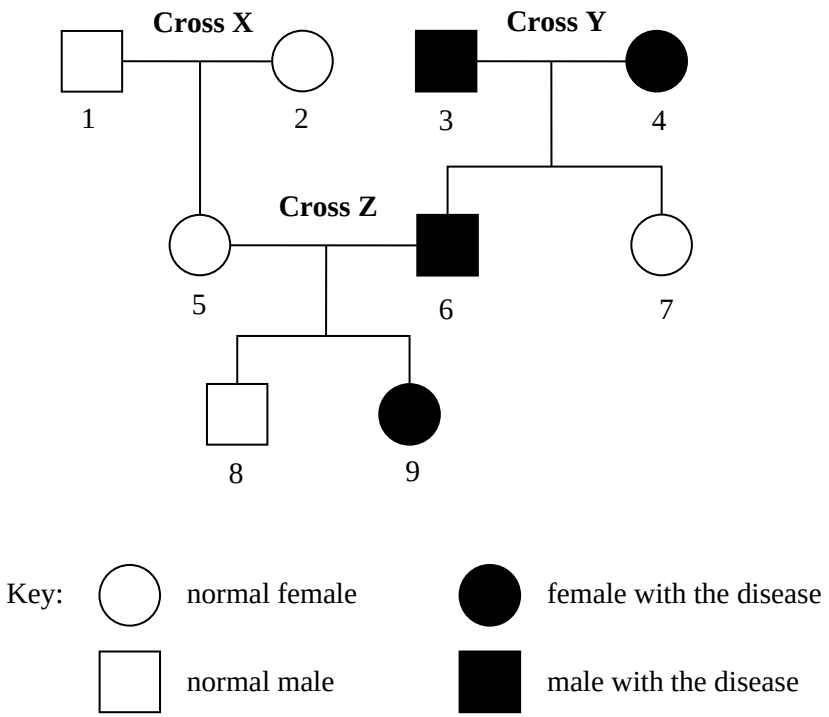
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	3
(c)	Presenting information	4

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|!|BCELB0260000013|!| ★

The pedigree below shows the inheritance a genetic disease caused by a dominant allele of a gene.



(a) State the cross that shows

(i) that the allele of the disease is ***not*** X-linked. (1 mark)

(ii) that the allele of the disease is ***not*** Y-linked. (1 mark)

(b) Deduce the genotype of individual 6. (Note: Marks will not be awarded for genetic diagrams.) (4 marks)

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(c) Individual 9 marries a man with the disease. Their first child is found to be normal. What is the probability of their second child being normal? Illustrate your answer with a genetic diagram. (5 marks)

##

(a) (i) Cross Y (1)

(ii) Cross Z (1)

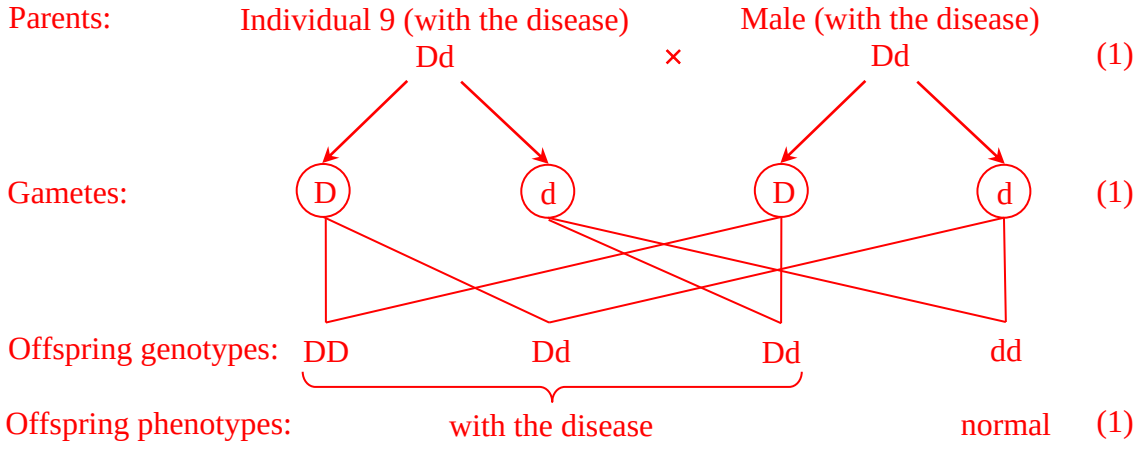
(b) Individual 6 has the disease. He must have at least one allele for the disease. (1)

Individual 8 is normal. He must be homozygous recessive. (1)

He must have received a recessive allele from individual 6. (1)

Hence, individual 6 is heterozygous. (1)

(c) Let D represent the allele for the disease and d represent the normal allele. (1)



The probability that their second child being normal = $^1/_4$ (or 25%) (1)

(11 marks)

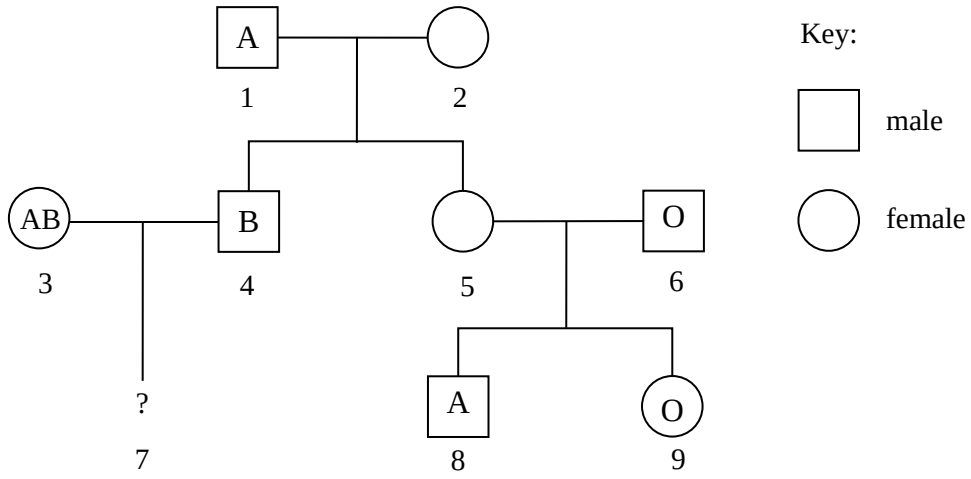
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##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	1
(b)	Analysing data	2
	Applying concepts	2
(c)	Presenting information	4
	Mathematical skills	1

|!|BCELB0260000014|!| ★

In humans, the ABO blood groups are determined by three alleles, I^A, I^B and i. Alleles I^A and I^B are co-dominant while allele i is recessive to both. The pedigree below shows the inheritance of ABO blood groups in a family.



(a) State the possible genotype(s) of

(i) individual 5. (1 mark)

(ii) individual 2. (1 mark)

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(b) By means of a genetic diagram, show the possible blood group(s) of individual 7. (4 marks)

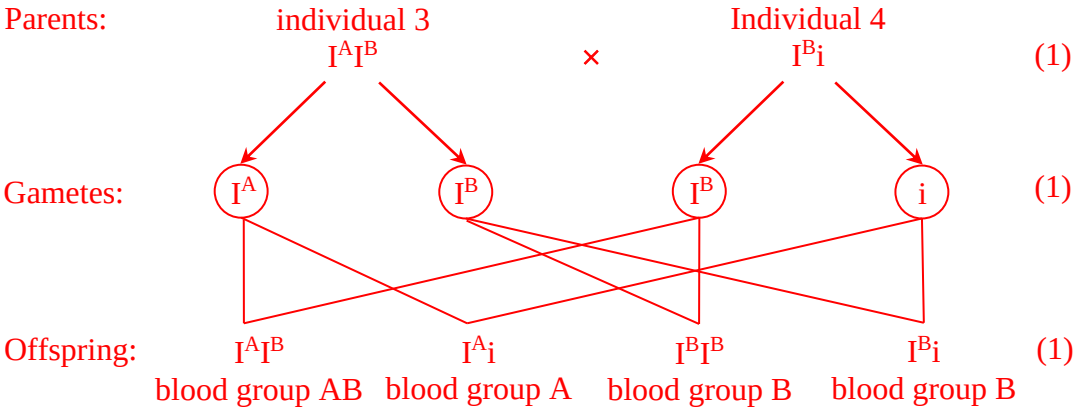
(c) In terms of antigens and antibodies, explain why individual 4 can donate blood to individual 3 without causing any ill effects. (3 marks)

##

(a) (i) I^Ai (1)

(ii) I^AI^B and I^Bi (1)

(b)



(1)

(1)

(1)

The possible blood groups of individual 7 are AB, A and B. (1)

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(c) Antigen B is present on the surface of the red blood cells of individual 4. (1)
Individual 3 do not have anti-B antibody in their plasma. (1)
Therefore, no/little agglutination will occur when individual 4's blood is transfused to individual 3. (1)

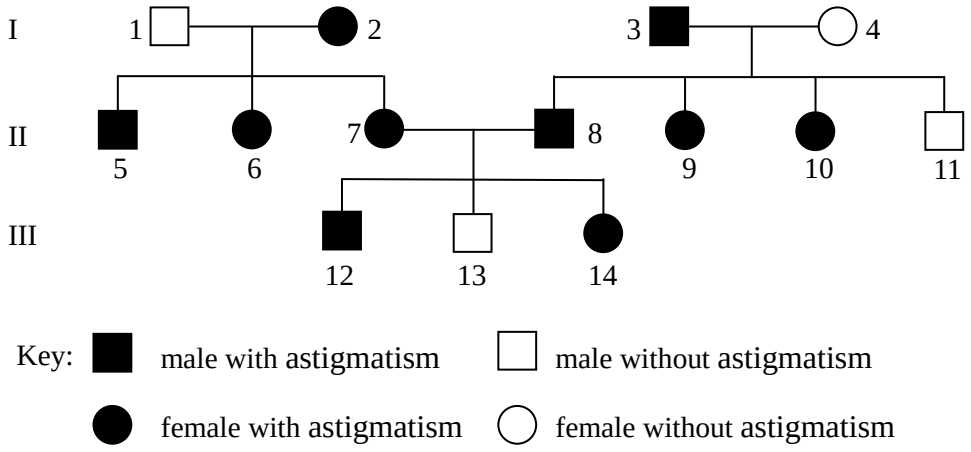
(9 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	1
(a)(ii)	Analysing data	1
(b)	Presenting information	4
(c)	Applying concepts	3

|!|BCELA0260000015|!| ★★

Heredity is one of the causes of astigmatism. The inheritance of this eye defect is controlled by a pair of alleles on an autosome (i.e. non-sex chromosome). The pedigree below shows the inheritance of astigmatism in a family.



(a) Based on the pedigree above, deduce whether the allele for astigmatism is dominant or recessive. (5 marks)

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(b) State the possible genotypes of individuals 2 and 3. Define the symbols you use.

(3 marks)

(c) Individuals 7 and 8 are going to have their fourth child. What is the probability of this child having normal vision? Illustrate your answer with a genetic diagram.

(4 marks)

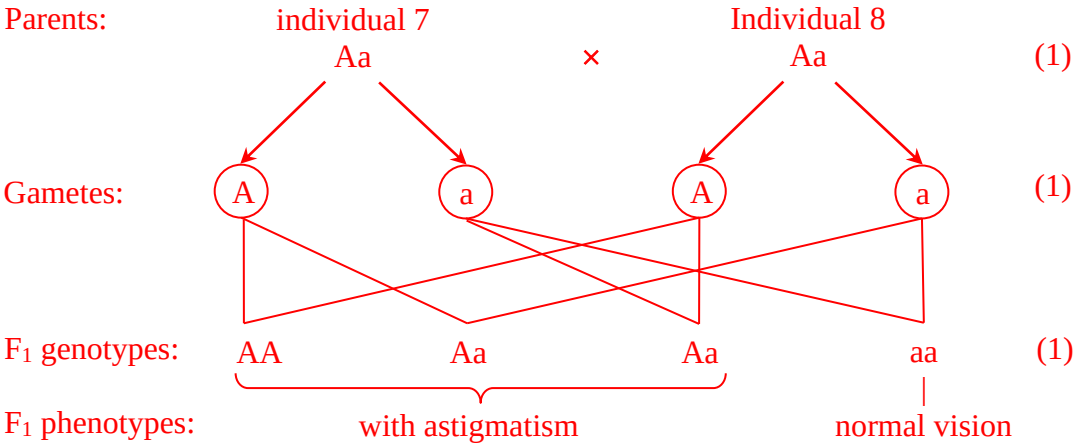
##

(a) Individual 13 does not have astigmatism. He must have at least one normal allele that is inherited from his parents, either individual 7 or 8. (1)
Both individuals 7 and 8 have astigmatism. They must have at least one allele for astigmatism. (1)
Hence, at least one of individuals 7 and 8 is heterozygous. (1)
In the heterozygous condition, only the dominant allele can be expressed. (1)
Thus, the allele for astigmatism is dominant. (1)

(b) Let A represent the allele for astigmatism, and a represent the allele for normal vision. (1)
Individual 2: AA, Aa (1)
Individual 3: Aa (1)

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(c)



Probability of the fourth child having normal vision = ¼ (or 25%) (1)

(12 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Applying concepts	3
(b)	Presenting information	1
	Analysing data	2
(c)	Presenting information	3
	Mathematical skills	1

|!|BCELA0260000016|!| ★★ **Cross topic**

In tulip plants, the petals can have markings called flecks. Whether the petals have flecks or not is controlled by a pair of alleles. In a study, two separate crosses were performed and the results are shown below.

Cross	Parent plants	Number of F_1 offspring	
		petals with flecks	petals without flecks
I	petals with flecks × petals without flecks	52	48
II	petals with flecks × petals without flecks	96	0

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(a) Which phenotype, petals with or without flecks, is caused by the dominant allele? Explain your answer without using a genetic diagram. (4 marks)

(b) State the genotypes of the parent plants with flecks in crosses I and II respectively. Define the symbols you use. (3 marks)

(c) The F_1 offspring with flecks produced from cross I were self-crossed. Use a genetic diagram to show the result. (3 marks)

(d) Is the cross mentioned in (c) a kind of sexual or asexual reproduction? Explain your answer. (2 marks)

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##

(a) Petals with flecks (1)

In cross II, some offspring must have received an allele for flecks and an allele for petals without flecks from their parents / must be heterozygous. (1)

All offspring have petals with flecks. (1)

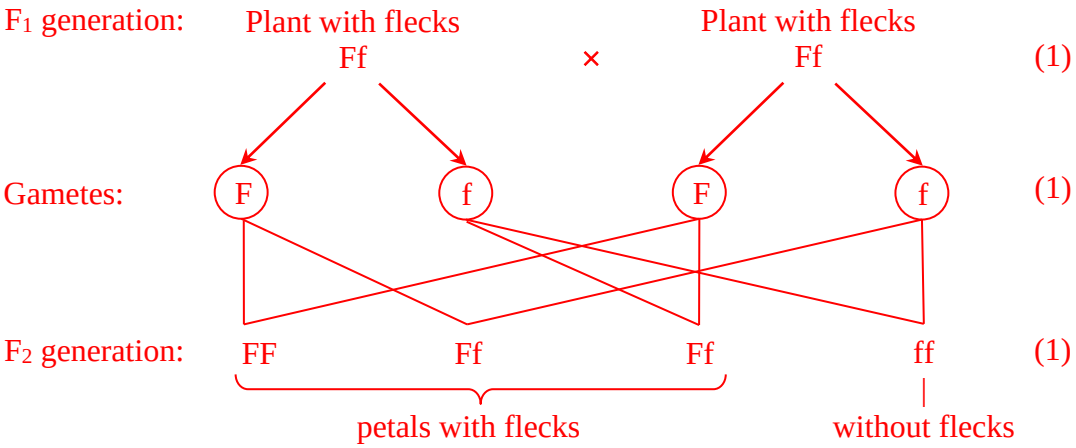
Therefore, the allele for flecks must have masked the other allele. (1)

(b) Let F represent allele for petals with flecks present, and f represent the allele for petals without flecks. (1)

Cross I: Ff (1)

Cross II: ff (1)

(c)



(1)

(1)

(1)

(d) Sexual reproduction (1)

because the self-cross involves the fusion of two gametes / fertilization. (1)

(12 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	3
(b)	Presenting information	1
	Analysing data	2
(c)	Presenting information	3
(d)	Understanding basic concepts	2

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|!|BCELA0260000017|!| ★★

In pea plants, the seed colour is controlled by a pair of allele. The allele for yellow seeds (Y) is dominant while the allele for green seeds (y) is recessive. In an investigation, four separate crosses were performed and the results are shown below.

Cross	**Parent plants grown from**	**Number of offspring (F_1 generation)**	
		yellow seeds	green seeds
A	yellow seed × yellow seed	88	0
B	yellow seed × yellow seed	66	21
C	yellow seed × green seed	50	42
D	yellow seed × green seed	93	0

(a) State all the possible genotypes of the parent plants in each cross. (4 marks)

Cross	**Genotypes of parent plants**
A	
B	
C	
D	

(b) If the yellow seeds produced from cross B were grown into plants and self-crossed, how many of these plants would produce green seeds in the F_2 generation? Illustrate your answer with a genetic diagram. (4 marks)

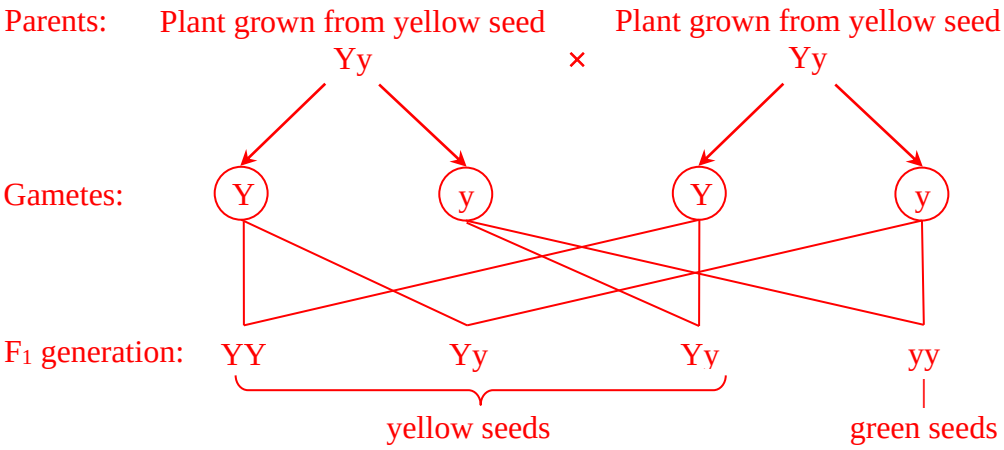
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##

(a) 1 mark for each cross correct (4)

Cross	Genotypes of parent plants
A	YY × YY ***OR*** YY × Yy
B	Yy × Yy
C	Yy × yy
D	YY × yy

(b) In cross B,



Only the heterozygous F_1 plants (Yy) would produce green seeds in the F_2 generation if self-crossed. (1)

The number of F_1 plants that would produce green seeds = $66 \times {}^{2}/_{3} = 44$ (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	4
(b)	Presenting information	2
	Applying concepts	1
	Mathematical skills	1

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|!|BCELB0260000018|!| ★

Albinism is a genetic disorder characterized by the lack of the pigment melanin in the body. The ability to produce melanin is controlled by a pair of alleles of a gene. Melanin is produced in both homozygous dominant and heterozygous individuals.

(a) Distinguish between the terms gene and allele. (2 marks)

(b) Explain the meaning of the term heterozygous. (1 mark)

(c) Is the allele for the production of melanin dominant or recessive? (1 mark)

(d) Use a genetic diagram to explain why an apparently normal couple could have a child with albinism. (4 marks)

(e) With reference to the genetic diagram in (d), predict the probability that the second child of the couple would have normal pigmentation. (1 mark)

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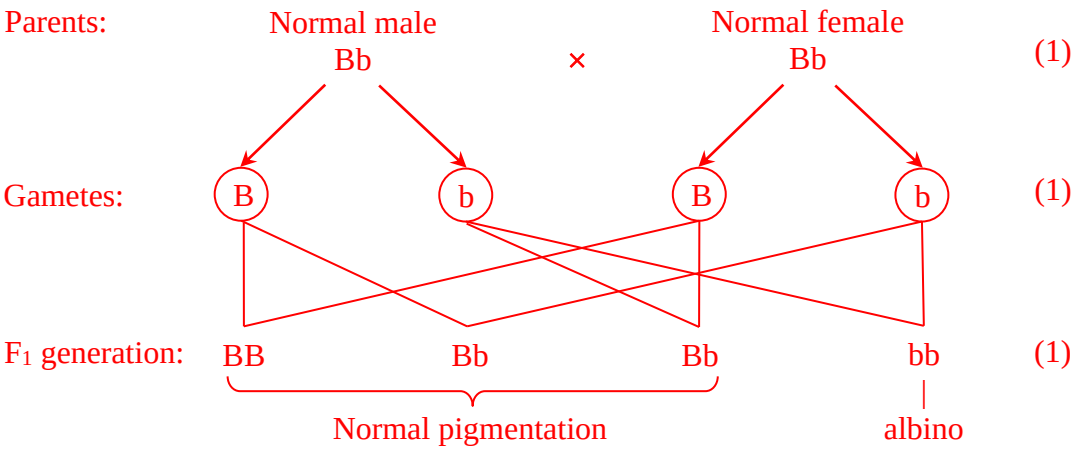
##

(a) A gene is a section of DNA / sequence of bases that codes for a polypeptide. (1)
An allele is an alternate form of a gene. (1)

(b) The presence of different alleles at a particular gene locus on homologous chromosomes (1)

(c) Dominant (1)

(d) Let B represent the dominant allele for melanin production, and b represent the recessive allele for albinism. (1)



(e) The probability of the second child having normal pigmentation = $^{3}/_{4}$ (or 75%) (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)	Analysing data	1
(d)	Presenting information	4
(e)	Mathematical skills	1

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|!|BCELB0260000019|!| ★

(a) Explain the following genetic terms.

(i) phenotype (1 mark)

(ii) allele (1 mark)

(iii) heterozygote (1 mark)

(b) The flower colour in a certain plant species is controlled by a pair of alleles. The table below shows the results of three separate crosses.

Cross	Parent plants	Number of offspring	
		red flower	white flowers
I	red flower × red flower	72	24
II	white flower × white flower	0	98
III	red flower × white flower	53	49

(i) Which of the crosses allows you to deduce which flower colour is dominant? (1 mark)

(ii) Without using genetic diagrams, deduce which flower colour is dominant. (5 marks)

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(iii) Using the symbol F to represent the dominant allele for flower colour and the symbol f to represent the recessive allele, give all the possible genotypes of

(1) the offspring with red flowers in cross I. (1 mark)

(2) the parent plants with red flowers in cross III. (1 mark)

##

(a) (i) Appearance / Expression of a character (1)

(ii) Different forms of a gene (1)

(iii) An organism with two different alleles of a gene (1)

(b) (i) Cross I (1)

(ii) Red flower is the dominant character. (1)

In cross I, the offspring with white flowers must have received at least one allele for white flowers that is inherited from the parent plants. (1)

Since the parent plants have red flowers, they must have at least one allele for red flowers. (1)

At least one of the parent plants is heterozygous. (1)

In the heterozygous condition, only the dominant allele can be expressed. (1)

(iii) (1) FF and Ff (1)

(2) Ff (1)

(11 marks)

##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(a)(iii)	Understanding basic concepts	1
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	2
	Applying concepts	3
(b)(iii)	Analysing data	2

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|!|BCELB0260000020|!| ★

In mice, the fur colour is controlled by a pair of alleles. A homozygous black mouse and a homologous brown mouse were allowed to mate. All the mice in the F_1 generation were black. The F_1 mice were then allowed to mate among themselves. The F_2 generation consisted of 74 black mice and 6 brown mice.

(a) Deduce which phenotype for fur colour is dominant. (3 marks)

(b) Using B to represent the dominant allele and b to represent the recessive allele, state the genotypes of the parental generation. (1 mark)

(c) Use a genetic diagram to show how the brown F_2 mice were produced from the F_1 mice. (3 marks)

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(d) How would you determine the genotype of a black F_2 mouse? Explain your answer. (3 marks)

##

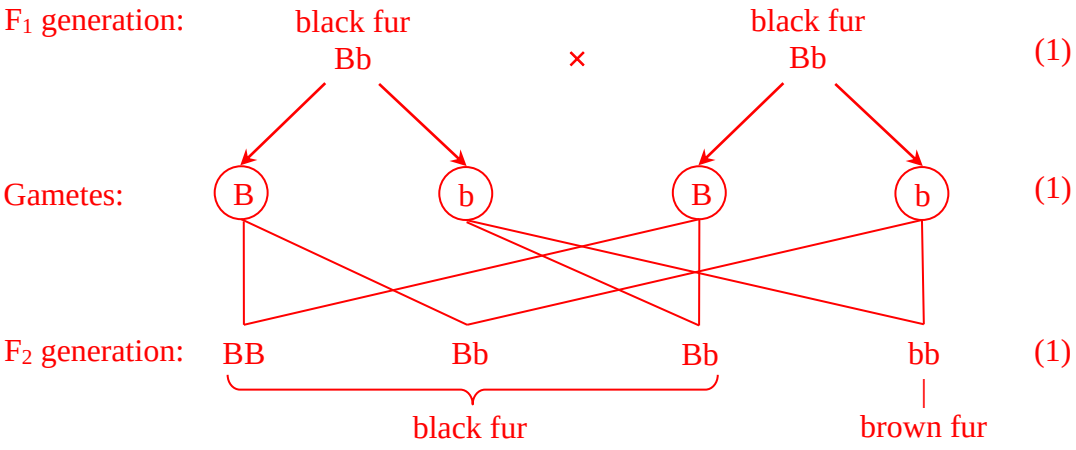
(a) Since both mice in the parental generation are homozygous, the F_1 mice produced must be heterozygous. (1)

In the heterozygous condition, the dominant allele will be expressed while the recessive allele will be masked. (1)

Therefore, blank fur is the dominant phenotype. (1)

(b) Black – BB; brown – bb (1)

(c)



(1) (1) (1)

(d) Cross the F_2 black mouse with a brown mouse. (1)

If the black mouse is homozygous (BB), all the offspring will be black. (1)

If the black mouse heterozygous (Bb), some offspring will be brown. (1)

(10 marks)

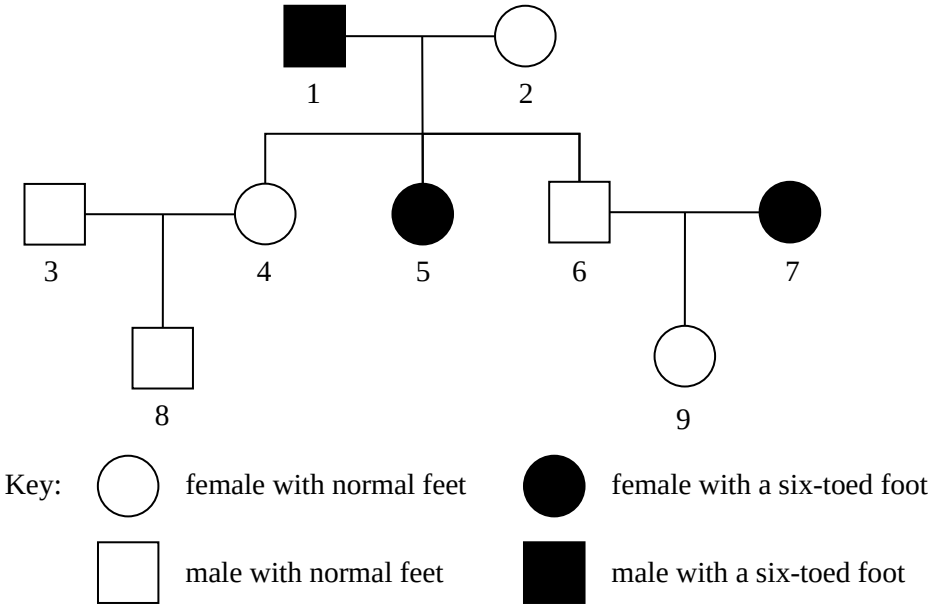
##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	1
(c)	Presenting information	3
(d)	Applying concepts	3

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|!|BCELA0260000021|!| ★★

The inheritance of six-toed feet is controlled by a pair of alleles. The pedigree below shows the inheritance of this trait in a family.



(a) After studying the pedigree, a student could not determine which trait is dominant but he made the following conclusion.

'Either individual 1 or individual 2 must be heterozygous.'

Do you agree with this conclusion? Explain your answer. (3 marks)

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(b) Given that the allele for a six-toed foot is dominant, deduce the possible genotype(s) of individual 5. (Note: Marks will not be awarded for genetic diagrams.) (4 marks)

(c) Individuals 6 and 7 are going to have another child. What is the probability of their second child having a six-toed foot? Illustrate your answer with a genetic diagram. (5 marks)

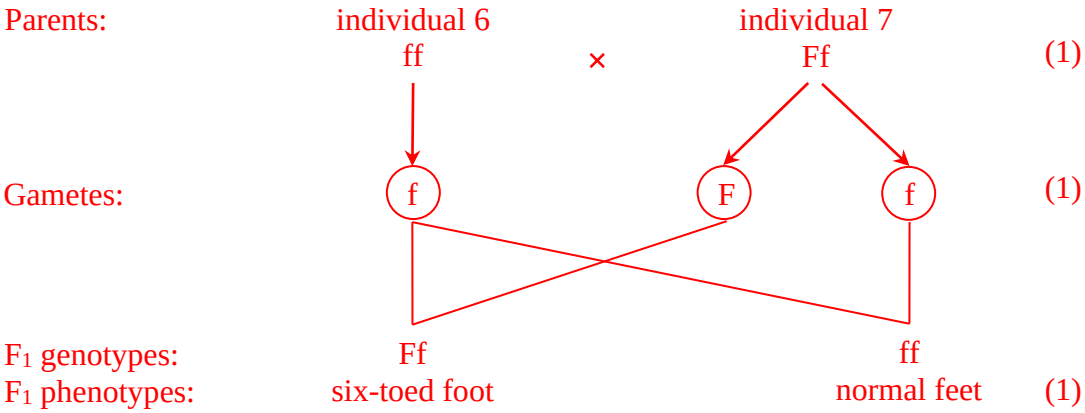
##

(a) Yes
Individuals 1 and 2 are of different phenotypes. Hence, one of them must be homozygous recessive. (1)
To produce offspring with different genotypes (1)
the other parent must be heterozygous, producing two types of gametes carrying different alleles. (1)
(Accept other reasonable answers.)

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(b) Individual 2 has normal feet. She must be homozygous recessive (1)
and has pass an allele for normal feet to individual 5. (1)
Individual 5 has a six-toed foot, she must have at least one allele for a six-toed foot. (1)
Hence, individual 5 is heterozygous. (1)

(c) Let F be the allele for a six-toed foot and f be the allele for normal feet. (1)



(1)
(1)
(1)

The probability for their second child to have a six-toed foot = $^1/_2$ (or 50%) (1)

(12 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	2
(b)	Analysing data	2
	Applying concepts	2
(c)	Presenting information	4
	Mathematical skills	1

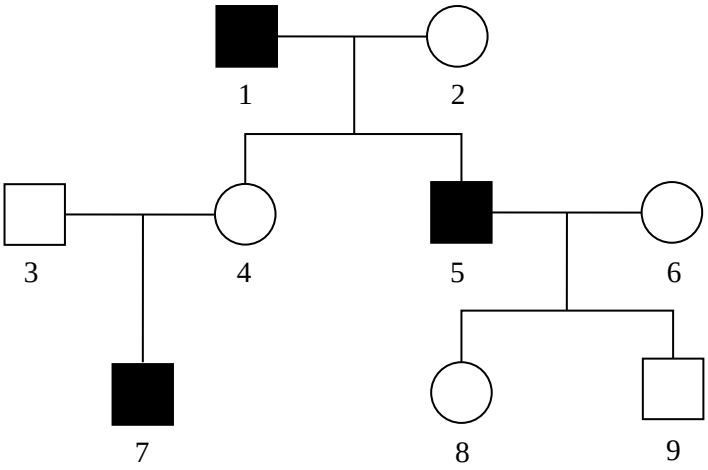
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|!|BCELB0260000022|!| ★

Haemophilia is a genetic disorder caused by a recessive allele of a gene located on the X chromosome.

(a) Are males or females more likely to have haemophilia? Explain your answer. (5 marks)

The pedigree below shows the inheritance of haemophilia in a family.



Key: normal male; male with haemophilia; normal female

(b) State all individuals in the pedigree who must be carriers of haemophilia. (2 marks)

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(c) Individuals 3 and 4 are going to have another child. What is the probability of this child being normal? Illustrate your answer with a genetic diagram. (5 marks)

##

(a) Males are more likely to have haemophilia. (1)

Females have two X chromosomes. (1)

A female will have haemophilia only if both of her X chromosomes carry the recessive allele. (1)

Males have one X chromosome and one Y chromosome. (1)

Since the gene is not found on the Y chromosome, a male having one recessive allele will have haemophilia. (1)

(b) Individuals 2, 4 and 8 (2)

(1 mark for two answers correct; deduct 1 mark for one wrong answer.)

(c) Let H be the normal dominant allele and h be the recessive allele for haemophilia. (1)

Parents: individual 3 normal male X^HY × individual 4 female carrier X^HX^h (1)

Gametes: X^H, Y, X^H, X^h (1)

F_1 genotypes:	X^HX^H	X^HX^h	X^HY	X^hY
F_1 phenotypes:	normal (female)	normal (carrier female)	normal (male)	haemophilic male

(1)

The probability of the second child being normal = $^3/_4$ (or 75%) (1)

(12 marks)

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##

	Skills assessed	Marks
(a)	Applying concepts	5
(b)	Analysing data	2
(c)	Presenting information	4
	Mathematical skills	1

|!|BCELA0260000023|!| ★★

In guinea pigs, fur colour is controlled by a gene. The allele for black fur (B) is dominant while the allele for white fur (b) is recessive. The length of fur is controlled by another gene located on a different chromosome. The allele for short fur (F) is dominant while the allele for long fur (f) is recessive.

(a) A pure-bred guinea pig with black short fur and a pure-bred guinea pig with white long fur were crossed.

(i) What is the term for this type of cross concerning the inheritance of two genes? (1 mark)

(ii) Draw a genetic diagram to show the result of this cross. (3 marks)

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(b) After the F_1 guinea pigs had grown up, they were mated with pure-bred guinea pigs with white long fur. Four types of offspring were produced in the F_2 generation: (1) black short fur, (2) black long fur, (3) white short fur, and (4) white long fur. Explain the result of the cross using Mendel's law of inheritance.
(Note: Marks will not be awarded for genetic diagrams.) (4 marks)

##

(a) (i) Dihybrid cross (1)

(ii)

Parents: pure-bred black short fur BBFF × pure-bred white long fur bbff (1)

Gametes: BF, bf (1)

F_1 genotype: BbFf
F_1 phenotype: black short fur (1)

(b) The pure-bred guinea pigs with white long fur will produce one type of gametes only. (1)
As the guinea pigs in F1 generation are heterozygous with genes for the two traits located on different chromosomes, the alleles for the traits are assorted independently. (1)
Therefore, four different types of gametes will be produced. (1)
After random fertilization, zygotes with four different combinations of genotypes will be produced, (1)
leading to the expressions of all four the possible phenotypes in the F_2 generation.

(8 marks)

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##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Presenting information	3
(b)	Applying concepts	4

|!|BCELA0260000024|!| ★★

In a breed of dogs, the fur can be brown or yellow. In addition, they can have a fluffy tail or a thin-haired tail. The two characteristics are controlled by two genes located on different chromosomes.

A pure-bred dog with brown fur and a fluffy tail and a pure-bred dog with yellow fur and a thin-haired tail were mated several times. All their offspring (F_1 generation) had brown fur and fluffy tails.

(a) Based on the cross between these two pure-bred dogs, deduce which phenotypes are dominant. (3 marks)

(b) After the F_1 generation had grown up, they were mated with pure-bred dogs with yellow fur and thin-haired tails. Four types of offspring were produced in the F_2 generation: (1) brown fur and fluffy tail, (2) brown fur and thin-haired tail, (3) yellow fur and fluffy tail, and (4) yellow fur and thin-haired tail.

What phenotypic ratio is expected in the F_2 generation? Illustrate your answer with a genetic diagram. (5 marks)

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##

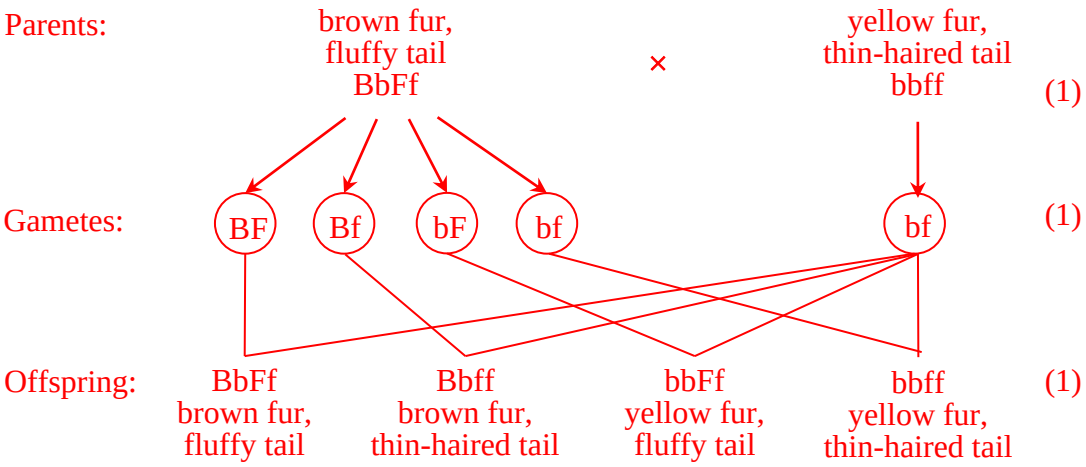
(a) Since both parents are pure-bred, the F_1 produced are all heterozygous. (1)

In the heterozygous condition, dominant alleles will be expressed while recessive alleles will be masked. (1)

Therefore, brown fur and fluffy tails are the dominant phenotypes. (1)

(b) Let B be the dominant allele for brown fur b be the recessive allele for yellow fur.

Let F be the dominant allele for a fluffy tail and f be the recessive allele for a thin-haired tail (1)



(1)

(1)

(1)

The phenotypic ratio in the F2 generation is expected to be 1:1:1:1 (1)

(8 marks)

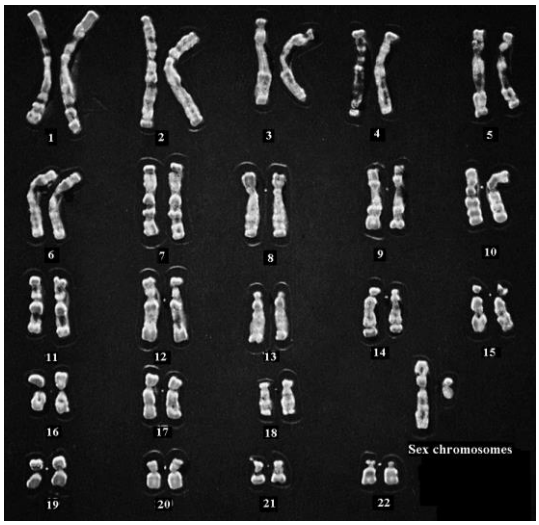
##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Presenting information	4
	Mathematical skills	1

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|!|BCELA0260000025|!| ★

The diagram below shows the karyotype of a person, T.



(a) Is T a male or a female? Explain your answer. (3 marks)

(b) Colour blindness is an X-linked recessive trait in humans. T is colour-blind while both parents of T are normal.

(i) What is the genotype of T's mother? (1 mark)

(ii) T's mother has recently given birth to a baby girl . What is the probability that this girl is colour-blind? Explain your answer. (Note: Marks will not be awarded for genetic diagrams.) (5 marks)

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##

(a) Person T is a male (1)

because the two sex chromosomes are different in size/length. (1)

and thus one is X chromosome and the other one is Y chromosome. (1)

(b) (i) heterozygous (1)

(ii) The probability is 0. (1)

T's father is a male and has only one X chromosome. (1)

Since he has normal colour vision, this X chromosome must bear the allele for normal colour vision. (1)

This X chromosome that bears the allele for normal colour vision must be inherited to his daughter. (1)

The allele for normal colour vision is dominant over the allele for colour blindness / will be expressed over the allele for colour blindness. (1)

Even if the girl receives an allele for colour blindness from the mother, the baby girl will have normal colour vision.

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	2
(b)(i)	Applying concepts	1
(b)(ii)	Applying concepts	5

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